

VENKATA VISHNU VARDHAN THOTA

Proddatur, Andhra Pradesh | +91 9182260519 | venkatavishnuvardhanthota@gmail.com

[linkedin.com/in/venkata-vishnu-varadhan-thota](https://www.linkedin.com/in/venkata-vishnu-varadhan-thota) | <https://vishnu-thota.vercel.app/> | github.com/Venkatavishnuvardhanthota

Final-year B.Tech student (AI & Data Science) seeking Data Analyst internships and entry-level roles, available immediately. Skilled in SQL (MySQL), Excel, Power BI, Tableau and Python for data cleaning, A/B testing, dashboards and sales forecasting. Completed 7 projects on 1M+ records, including a 588,101-record A/B test (0.77-point conversion lift, $p < 0.001$) and a MySQL-to-Power BI sales pipeline across 1,115 stores; 2 projects run as live apps. IBM Data Science and Google Cloud Data Analytics certified.

TECHNICAL SKILLS

- **Data Analysis & BI:** SQL (MySQL), Microsoft Excel (Pivot Tables, Power Query, VLOOKUP, XLOOKUP), Power BI (DAX), Tableau, Pandas, NumPy, Matplotlib, Seaborn, Plotly
- **Core Analytics:** EDA, Data Cleaning, Data Wrangling, Dashboard Development, Business Intelligence, Hypothesis Testing, A/B Testing, SQL Window Functions, KPI Reporting
- **Machine Learning & NLP:** Scikit-learn, Logistic Regression, Random Forest, Feature Engineering, Model Evaluation (AUC-ROC, Cross-Validation), SMOTE, NLTK, VADER, TextBlob, TF-IDF, WordCloud
- **Tools:** Git, GitHub, Streamlit, Jupyter Notebook
- **Programming Languages:** Python, Java

PROJECTS

Marketing A/B Testing Analysis | Python, SQL (MySQL), Tableau, Git, GitHub

- Built a MySQL-to-Python pipeline for **588,101 marketing records**, resolving a silent import-truncation failure via staged **LOAD DATA INFILE** transformation and **SQL window functions** for hourly/daily trend aggregation.
- Applied a **two-proportion Z-test** to validate a **0.77-point conversion lift ($p < 0.001$, 95% CI: 0.60%–0.94%)** over a PSA control group, translating results into an estimated **₹21.7 lakh revenue impact** via an interactive Tableau dashboard.

Rossmann Store Sales Forecasting Pipeline | Python, SQL, Pandas, NumPy, Prophet, XGBoost, Scikit-learn, Power BI, DAX, Excel

- Built a MySQL data pipeline processing **1M+ records** across **1,115 stores** using **joins**, **CTEs**, and **window functions** across seven analytical queries, identifying and resolving a **silent NULL-handling bug** that was fabricating data values.
- Benchmarked **Prophet** against **XGBoost** after tuning Prophet through **baseline**, **holiday-aware**, and **seasonality stages**, then selected **XGBoost** as the production model after **RandomizedSearchCV** tuning cut forecast error from **21.00% to 17.13% MAPE**, retaining Prophet for its **interpretable trend and seasonality decomposition**.

Customer Churn Prediction System | Python, Scikit-learn, Pandas, SMOTE, Streamlit, Git, GitHub

- Addressed a **73.5% / 26.5%** class imbalance by applying **SMOTE exclusively to the training dataset**, expanding training samples from **5,625 to 8,260**, then trained Logistic Regression and Random Forest models with Random Forest achieving an **AUC-ROC of 0.8343**.
- Identified that **month-to-month customers experienced 42.7% churn compared to only 2.9% for two-year contracts**, providing actionable retention insights through exploratory data analysis and feature importance evaluation.
- Deployed as a live web app on Streamlit Cloud — customer-churn-prediction-0001.streamlit.app

Product Review Sentiment Dashboard | Python, NLTK, VADER, TextBlob, TF-IDF, WordCloud, Plotly, Scikit-learn, Git, GitHub

- Built an end-to-end NLP sentiment analysis pipeline processing **568,454 Amazon Fine Food Reviews (1999–2012)** using **VADER**, **TextBlob**, and custom preprocessing, achieving **79.6% agreement** with ground-truth star ratings while extracting high-impact negative phrases using **TF-IDF**.
- Analyzed sentiment misclassification patterns and discovered that nearly **50% of one-star reviews** were classified as **POSITIVE** by VADER due to expectation-framing language, recommending a hybrid sentiment scoring approach for improved production reliability.
- Deployed as a live web app on Streamlit Cloud — prduct-sentiment-dashboard.streamlit.app

EDUCATION

B.Tech, Artificial Intelligence & Data Science — Study World College of Engineering | 2024–2027 | CGPA 7.72

Diploma, Computer Science — Govt. Polytechnic College | 2021–2024 | CGPA 8.10

CERTIFICATIONS

- [Google Cloud Data Analytics Certificate](#) — Google
- [IBM Data Science Professional Certificate](#) — Coursera
- Oracle Databases for Developers: Foundations — Oracle

ACHIEVEMENTS

- Earned **2★ SQL Badge** on HackerRank by solving SQL programming and database query challenges.